

Detecting Anomalous DNS Activity using Zeek, TryHackMe

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Objective

The purpose of this analysis is to investigate a triggered alert titled 'Anomalous DNS Activity'. Using Zeek, the task is to inspect a provided PCAP file (dns-tunneling.pcap) and confirm whether the alert is a true positive. Indicators of DNS tunneling and anomalous traffic will be identified using Zeek logs and filters.

Tools & Environment

- Zeek (Network Security Monitor)
- Ubuntu Virtual Machine
- TryHackMe (Anomalous DNS lab environment)
- Command-line utilities: cat, zeekcut, sort, uniq, head

Scenario Recap

A DNS alert was triggered suggesting possible DNS tunneling — a technique often used by attackers to exfiltrate data via DNS queries. The PCAP file dns-tunneling.pcap was provided for deep packet analysis.

Path to PCAP:
cd Desktop/Exercise-Files/anomalous-dns

Zeek command used:
zeek -C -r dns-tunneling.pcap

Methodology & Commands Used

STEP	DESCRIPTION	COMMAND
1	Inspect DNS log	cat dns.log

2	Preview first 10 lines	cat dns.log head -n 10
3	Count DNS query types	cat dns.log zeekcut qtype_name sort uniq -c
4	Inspect conn.log	cat conn.log
5	Find longest connection	cat conn.log zeek-cut duration sort uniq
6	Identify involved hosts	cat conn.log zeek-cut id.orig_h sort uniq

Key Findings

1. High Volume of DNS AAAA Queries

- Found: 320 records of AAAA (IPv6) DNS query types
- This is suspicious, especially since the source IP is IPv4

2. Unusually Long Connection

- Maximum connection duration: 9.420791 seconds
- Could indicate persistent communication

3. Internal Host Identified

- IP: 10.20.57.3
- Initiated excessive DNS queries, all to the same external domain

4. True Positive Confirmed

- Volume + Pattern + Destination consistency confirms DNS tunneling behavior

Indicators of Compromise (IOCs)

TYPE	VALUE
SOURCE IP	10.20.57.3
QUERY TYPE	AAAA (320 instances)
LONGEST CONNECTION	9.420791s
PATTERN	Repetitive DNS queries to the same domain

Analysis Summary

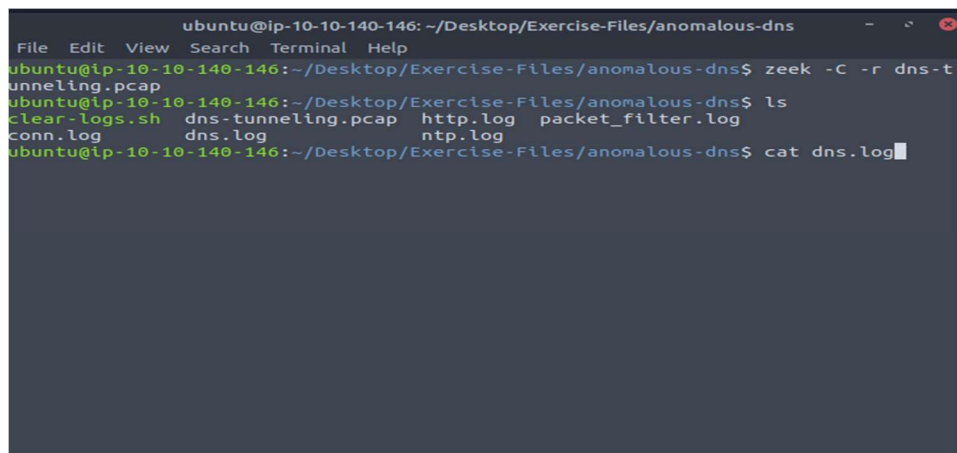
Zeek was effective in uncovering suspicious DNS activity indicative of DNS tunneling, a known exfiltration method. The consistent use of AAAA queries by a single internal host, along with the prolonged connection durations, raises a red flag.

Recommendations

- Monitor all future AAAA query spikes from internal IPv4-only hosts
- Implement deep packet inspection or DNS proxy filtering
- Consider adding thresholds or alerts in SIEM based on similar patterns

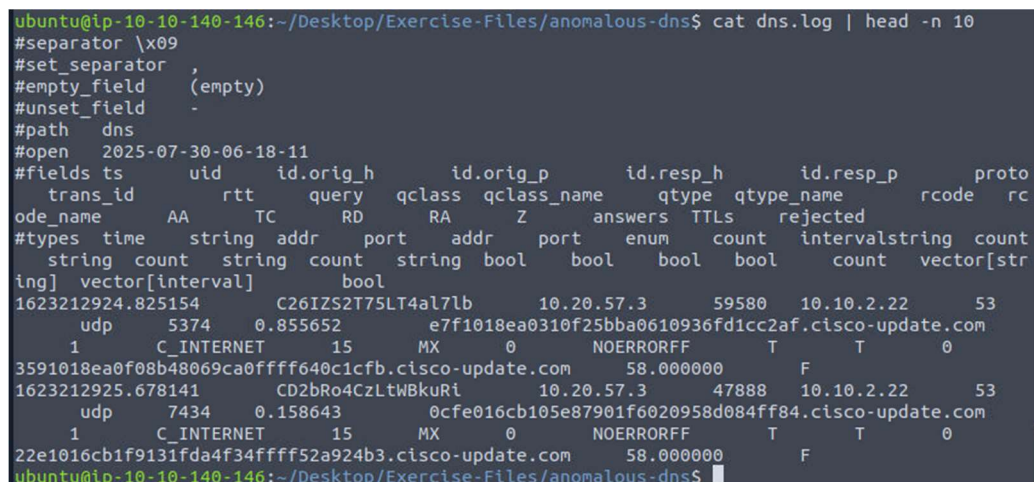
Screenshots

- Screenshot 1: Zeek command run on PCAP

A terminal window showing the execution of Zeek commands on a PCAP file. The user runs 'zeek -C -r dns-tunneling.pcap', lists files, and then cat's the generated 'dns.log' file.

```
ubuntu@ip-10-10-140-146: ~/Desktop/Exercise-Files/anomalous-dns
File Edit View Search Terminal Help
ubuntu@ip-10-10-140-146:~/Desktop/Exercise-Files/anomalous-dns$ zeek -C -r dns-tunneling.pcap
ubuntu@ip-10-10-140-146:~/Desktop/Exercise-Files/anomalous-dns$ ls
clear-logs.sh  dns-tunneling.pcap  http.log  packet_filter.log
conn.log       dns.log             ntp.log
ubuntu@ip-10-10-140-146:~/Desktop/Exercise-Files/anomalous-dns$ cat dns.log
```

- Screenshot 2: Output of dns.log

A terminal window showing the first 10 lines of the 'dns.log' file. The output is a structured log entry with various fields like trans_id, rtt, query, qclass, qclass_name, qtype, qtype_name, rcode, and rc.

```
ubuntu@ip-10-10-140-146:~/Desktop/Exercise-Files/anomalous-dns$ cat dns.log | head -n 10
#separator \x09
#set_separator ,
#empty_field (empty)
#unset_field -
#path dns
#open 2025-07-30-06-18-11
#fields ts uid id.orig_h id.orig_p id.resp_h id.resp_p proto
trans_id rtt query qclass qclass_name qtype qtype_name rcode rc
ode_name AA TC RD RA Z answers TTLS rejected
#types time string addr port addr port enum count intervalstring count
ing] vector[interval] bool
1623212924.825154 C26IZS2T75LT4al7lb 10.20.57.3 59580 10.10.2.22 53
udp 5374 0.855652 e7f1018ea0310f25bba0610936fd1cc2af.cisco-update.com
1 C_INTERNET 15 MX 0 NOERRORFF T T 0
3591018ea0f08b48069ca0ffff640c1cfb.cisco-update.com 58.000000 F
1623212925.678141 CD2bRo4CzLtWBkuRi 10.20.57.3 47888 10.10.2.22 53
udp 7434 0.158643 0cfe016cb105e87901f6020958d084ff84.cisco-update.com
1 C_INTERNET 15 MX 0 NOERRORFF T T 0
22e1016cb1f9131fda4f34ffff52a924b3.cisco-update.com 58.000000 F
ubuntu@ip-10-10-140-146:~/Desktop/Exercise-Files/anomalous-dns$
```

```
ubuntu@ip-10-10-53-244:~/Desktop/Exercise-Files/anomalous-dns$ cat dns.log | zee
k-cut qtype_name | sort | uniq -c
    11 A
   320 AAAA
  2232 CNAME
  2314 MX
    23 PTR
  2347 TXT
ubuntu@ip-10-10-53-244:~/Desktop/Exercise-Files/anomalous-dns$
```

```
ubuntu@ip-10-10-53-244:~/Desktop/Exercise-Files/anomalous-dns$ cat conn.log | head -n 10
#separator \x09
#set_separator ,
#empty_field (empty)
#unset_field -
#path conn
#open 2025-07-30-06-33-19
#fields ts uid id.orig_h id.orig_p id.resp_h id.resp_p
p proto service duration orig_bytes resp_bytes conn_status
te local_orig local_resp missed_bytes history orig_pkts o
orig_ip_bytes resp_pkts resp_ip_bytes tunnel_parents
#types time string addr port addr port enum string interval
count count string bool bool count string count count c
count set[string]
1623212924.825154 CH3n6F3H8DzwdX0smi 10.20.57.3 59580 10.10.2.
22 53 udp dns 0.855652 80 175 SF - -
0 Dd 1 108 1 203 - -
1623212925.678141 CTzNeiZTb8CNmmwXh 10.20.57.3 47888 10.10.2.
22 53 udp dns 0.158643 80 175 SF - -
0 Dd 1 108 1 203 - -
ubuntu@ip-10-10-53-244:~/Desktop/Exercise-Files/anomalous-dns$
```

- Screenshot 5: Internal IP output

[illegible]